

Fluid Mechanics & Hydraulic Machinery - 4022

Diploma in Mechanical Engineering / Manufacturing Technology. Semester 4. Credits 4.
Program Core. Periods: 4 per week, 60 per semester.

This PDF notes file is generated from the 4022 handbook lesson. It is a compact printable companion containing official overview, roadmap, module chapters, formulas, solved examples, model question paper and answer points.

Course objectives:

- Identify properties of fluids and pressure measurement techniques.
- Understand fluid flow and its engineering applications.
- Find losses in pipe flow.
- Understand construction, working and performance testing of turbines and pumps.

Course outcomes:

CO1: Explain fluid properties and pressure measurement techniques - 12 hours - Understanding - PO1=3, PO2=3, PO7=1.

CO2: Apply conservation laws to notches, pipes and orifices - 17 hours - Applying - PO1=3, PO2=3, PO7=1.

CO3: Describe construction, working and testing of turbines - 17 hours - Understanding - PO1=3, PO2=3, PO7=1.

CO4: Describe construction, working and testing of pumps - 12 hours - Understanding - PO1=3, PO2=3, PO7=1.

Prerequisites: Mathematics I and II for problem solving; Applied Physics I and II for physics of fluids.

Module 1 - Fluid Properties and Pressure Measurement

Core topics: Density, specific gravity, specific weight, specific volume, dynamic viscosity, kinematic viscosity, surface tension, capillarity, vapour pressure and compressibility.

Density is mass per unit volume. Specific weight is weight per unit volume. Specific volume is volume per unit mass. Specific gravity is density ratio with water and has no unit.

Viscosity is resistance to relative motion between fluid layers. Dynamic viscosity is force effect; kinematic viscosity is dynamic viscosity divided by density.

Pressure is force per unit area. Pressure head is pressure expressed as height of fluid column. Absolute pressure is measured from perfect vacuum. Gauge pressure is measured from atmospheric pressure. Vacuum pressure is below atmospheric pressure.

Pascal law: pressure applied to a confined fluid is transmitted equally in all directions. Hydraulic press, hydraulic lift and hydraulic brakes use this principle.

Pressure instruments: Piezometer for simple positive pressure, U-tube manometer, differential manometer, inverted differential manometer and Bourdon pressure gauge.

Formulas: $\rho = m/V$; $\gamma = \rho \cdot g$; $SG = \rho_{\text{fluid}} / \rho_{\text{water}}$; $p = F/A$; $h = p / (\rho \cdot g)$.

Solved example: If pressure is 196.2 kPa in water, $h = p / (\rho \cdot g) = 196200 / (1000 \cdot 9.81) = 20$ m of water.

Exam tip: Draw pressure scale and instrument diagram before explanation. Mention units in all numerical answers.

Module 2 - Fluid Flow, Bernoulli, Notches and Pipe Losses

Core topics: steady/unsteady, uniform/non-uniform, compressible/incompressible, rotational/irrotational and one-dimensional/multidimensional flows.

Discharge Q is volume flow rate. Streamline is tangent to velocity direction. Path line is actual path followed by a fluid particle.

Continuity equation for steady incompressible flow: $A_1 \cdot V_1 = A_2 \cdot V_2 = Q$. It represents conservation of mass.

Bernoulli equation: $p/\gamma + V^2/(2g) + z = \text{constant}$. It represents conservation of pressure head, velocity head and datum head along a streamline.

Limitations: steady flow, incompressible fluid, inviscid assumption, along streamline, no pump/turbine work between sections and no major energy loss.

Applications: Venturimeter, orifice meter and Pitot tube. Notches and orifices measure discharge. Rectangular and triangular notches are important.

Pipe losses: Major loss is friction loss. Minor losses occur at entry, exit, valves, bends, sudden enlargement and sudden contraction. HGL and TEL show energy variation.

Formulas: $Q = AV$; $A_1 V_1 = A_2 V_2$; $h_f = 4fLV^2/(2gd)$; $h_m = K V^2/(2g)$.

Solved example: If $A_1 = 0.04 \text{ m}^2$, $A_2 = 0.02 \text{ m}^2$ and $V_1 = 1.5 \text{ m/s}$, $V_2 = A_1 V_1 / A_2 = 3 \text{ m/s}$ and $Q = 0.06 \text{ m}^3/\text{s}$.

Module 3 - Impact of Jets and Hydraulic Turbines

A water jet has momentum. When jet strikes a plate or vane, change in momentum produces force and work. Turbines extract energy by changing water momentum and pressure.

Hydroelectric plant layout: reservoir, dam, intake, trash rack, penstock, surge tank, turbine, draft tube, generator, tailrace and switchyard.

Turbine classification: impulse or reaction; tangential, radial, axial or mixed flow; high head, medium head or low head.

Pelton wheel: impulse turbine for high head and low discharge. Francis turbine: reaction turbine for medium head. Kaplan turbine: axial reaction turbine for low head and high discharge.

Draft tube recovers kinetic energy at reaction turbine outlet and discharges water to tailrace. Cavitation occurs when local pressure falls below vapour pressure; bubbles collapse and damage surfaces.

Formulas: water power= $\rho \cdot g \cdot Q \cdot H$. Turbine efficiency=output power/water power. Unit quantities help compare turbines under unit head.

Solved example: $Q=0.08 \text{ m}^3/\text{s}$, $H=40 \text{ m}$. Water power= $1000 \cdot 9.81 \cdot 0.08 \cdot 40=31.39 \text{ kW}$. If efficiency is 75 percent, output power= 23.54 kW .

Exam tip: For turbine answers, draw layout or runner sketch and then explain water energy conversion.

Module 4 - Hydraulic Pumps

Centrifugal pump: rotating impeller gives kinetic energy to water; casing converts part of kinetic energy into pressure energy. Main parts: impeller, casing, suction pipe, foot valve, strainer, delivery pipe and shaft.

Priming: filling pump casing and suction pipe with water before starting. Without priming, air prevents proper suction in centrifugal pump.

Cavitation: formation and collapse of vapour bubbles due to low pressure. Effects: noise, vibration, pitting, reduced efficiency and damage.

Manometric head: head against which pump works. Manometric efficiency relates useful head to impeller energy. Overall efficiency compares water power and shaft power.

Reciprocating pump: piston/plunger produces suction and delivery. Single acting pump delivers in one stroke; double acting pump delivers in both strokes.

Slip: theoretical discharge minus actual discharge. Negative slip occurs when actual discharge exceeds theoretical under special conditions.

Special pumps: hydraulic ram, air lift pump, jet pump, monoblock coupled submersible pump, propeller pump and turbine pump.

Formulas: Water power= $\rho \cdot g \cdot Q \cdot H$. Overall efficiency=water power/shaft power. Q single acting= $A \cdot L \cdot N / 60$. Slip= $Q_{th} - Q_{act}$.

Solved example: $Q=0.05 \text{ m}^3/\text{s}$, $H=30 \text{ m}$, shaft power=22 kW. Water power=14.715 kW. Efficiency= $14.715/22=66.9$ percent.

Formula Bank and Solved Examples

Formula bank:

1. Density: $\rho = m/V$, unit kg/m^3 . Used for property and mass calculations.
2. Specific weight: $\gamma = \rho \cdot g$, unit N/m^3 . Used in pressure head relation.
3. Specific gravity: $SG = \rho_{\text{fluid}} / \rho_{\text{water}}$, no unit.
4. Pressure: $p = F/A$, unit Pa .
5. Pressure head: $h = p / (\rho \cdot g)$, unit m of fluid.
6. Discharge: $Q = A \cdot V$, unit m^3/s .
7. Bernoulli: $p/\gamma + V^2/(2g) + z = \text{constant}$.
8. Darcy loss: $h_f = 4fLV^2/(2gd)$.
9. Minor loss: $h_m = K V^2/(2g)$.
10. Hydraulic power: $P = \rho \cdot g \cdot Q \cdot H$.
11. Pump efficiency: $\eta = \text{water power} / \text{shaft power}$.

Worked example 1: A liquid has mass 600 kg and volume 0.75 m^3 . Density = 800 kg/m^3 . Specific weight = 7848 N/m^3 .

Worked example 2: A pipe reduces from 200 mm to 100 mm. Velocity in large pipe is 1.2 m/s . $V_2 = (D_1/D_2)^2 \cdot V_1 = 4.8 \text{ m/s}$. $Q = 0.0377 \text{ m}^3/\text{s}$.

Common mistakes: skipping units, using diameter instead of area, forgetting coefficient of discharge, drawing unlabelled diagrams and mixing pump/turbine efficiencies.

Practice, Expected Questions and Model Paper

Expected questions:

- Define density, specific weight, specific volume, specific gravity and viscosity.
- State and explain Pascal law with application.
- Explain piezometer, U-tube manometer, differential manometer and Bourdon gauge.
- Classify fluid flow and explain stream line and path line.
- State continuity equation and Bernoulli equation with limitations.
- Explain Venturimeter, orifice meter and Pitot tube.
- Explain rectangular and triangular notches and hydraulic coefficients of orifice.
- Explain Darcy equation, Chezy equation, HGL and TEL.
- Explain impact of jet on fixed/moving plates and curved vanes.
- Draw hydroelectric power plant layout and explain turbine selection.
- Explain Pelton, Francis, Kaplan turbine, draft tube and cavitation.
- Explain centrifugal pump, priming, cavitation, reciprocating pump, slip and special pumps.

Model paper outline: Part A - 10 short answers. Part B - 8 paragraph/diagram questions. Part C - 4 descriptive/numerical questions. Cover all four modules. Fresh questions should not copy official questions.

Practice objective answers: Density is mass/volume. Venturimeter works on Bernoulli theorem. Pelton is high head turbine. Priming is required in centrifugal pump. Cavitation occurs when pressure falls below vapour pressure.

Complete Model Answers and References

Model answer numerical 1: $m=720 \text{ kg}$, $V=0.9 \text{ m}^3$. $\rho=m/V=800 \text{ kg/m}^3$. $\gamma=\rho \cdot g=7848 \text{ N/m}^3$. $SG=800/1000=0.8$.

Model answer numerical 2: $D_1=0.2 \text{ m}$, $D_2=0.1 \text{ m}$, $V_1=1.2 \text{ m/s}$. $A_1/A_2=4$. $V_2=4.8 \text{ m/s}$.
 $Q=\pi/4 \cdot 0.2^2 \cdot 1.2=0.0377 \text{ m}^3/\text{s}$.

Model answer numerical 3: Pump $Q=0.05 \text{ m}^3/\text{s}$, $H=30 \text{ m}$, shaft power= 22 kW . Water power= $1000 \cdot 9.81 \cdot 0.05 \cdot 30=14.715 \text{ kW}$. Overall efficiency= 66.9 percent .

Answer writing format: definition, labelled diagram, working principle, formula, application, conclusion. For numerical answers: given data, formula, substitution, calculation, final answer with unit.

References from supplied syllabus:

- Fluid Mechanics and Hydraulic Machines - Dr. R.K. Bansal.
- Hydraulic, Fluid Mechanics and Hydraulic Machines - R.S. Khurmi.
- Hydraulics and Fluid Mechanics - Dr. P.N. Modi and Dr. S.M. Seth.
- Hydraulics, Fluid Mechanics and Fluid Machines - S. Ramamurtham.
- Fluid Mechanics - Yunus A. Cengel and John M. Cimbala.

Online resources from syllabus: Elsevier Fluid Mechanics, Cambridge Introduction to Fluid Dynamics and NPTEL course 112105183.